

F-crystals and the Hodge filtration

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Newton-above - Hodge

Let k perfect field char $p > 0$.

σ G $W = W(k)$ with vector

X/k smooth proper variety,

free finite rank W -module

$\downarrow$

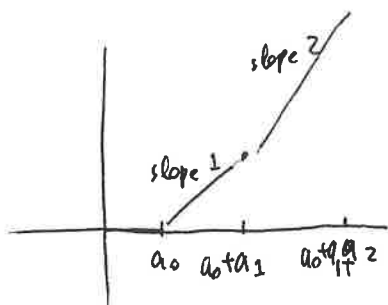
$$M^n := H_{\text{cris}}^n(X/W)_{\text{tf}}$$

$\hat{=}$ torsion-free quotient

$$\Phi: M^n \rightarrow M^n \quad \sigma\text{-semi-linear Frob.}$$

"F-crystal structure"

Def. For a sequence of nonnegative integers $(a_0, \dots, a_n, \dots)$, its Hodge polygon is its polygon



3 polygons. ① (geometric) Hodge polygon HP_{geom}^n def'd by $(h_0, \dots, h_n)$,

$$h_i = \dim H^{n-i}(X, \Omega_{X/k}^i)$$

② (Frobenius) Hodge polygon def'd by elementary divisors $M^n / \Phi(\sigma^* M^n) \simeq \bigoplus_{j=1}^{\text{rk } M} W / P^{a_j} W$

$$b_j = \# \{i : a_i = j\}, \quad HP_F^n \rightsquigarrow (b_0, \dots, b_n, \dots)$$

$$a_1 \leq \dots \leq a_{\text{rk } M}$$

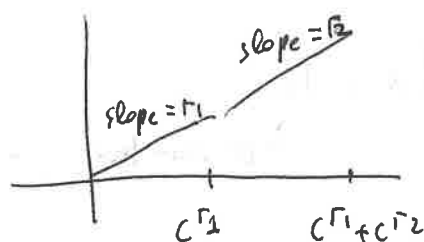
③ Newton polygon of $\Phi: M^n \rightarrow M^n$ NP_Φ

choose basis $e_1, \dots, e_{rk M}$ of M^n ,

$$\Phi = \begin{pmatrix} T(e_1) & \dots & T(e_n) \end{pmatrix} \quad \text{eigenvalues } \lambda_1 \dots \lambda_{rk M}$$

ordering the valuations $r_1 \leq \dots \leq r_m$

$$C^r = \# \{ \lambda_i : v_p(\lambda_i) = r \}$$



Th. (Mazur, Ogus, Katz's conj.)

NP_Φ lies above $HP_{geom.}$
($\geq$)

In fact, $NP_\Phi \geq HP_F \geq HP_{geom.}$
 | |
 easy hard
 (pure linear algebra)

Pure linear algebra

Choosing a matrix for Φ
idea. Newton polygon coincides w/ NP of the char. poly. of Φ .

$$X^n + a_1 X^{n-1} + \dots + a_n$$

Let $e^0, e^1, \dots$ denote the Hodge numbers (j such that $b_j \neq 0$)

Passing to $\Lambda^i \Phi: \Lambda^i \otimes^* M \rightarrow \Lambda^i M$, this is divisible by $P^{E(i)}$ (where $(i, E(i))$ lies on HP_F)

statement reduces to $v_p(a_i) \geq E(i)$. true by def'n of NP.

Note. Th doesn't assume X lifts (even formally) to $W(k)$, but some ingredients do.

First ingredient: purely local result.

Assume X lifts to $\tilde{X}/W(k)$

Consider $\Omega_{\tilde{X}/W(k)}^\bullet$

Define $\eta_p \Omega_{\tilde{X}/W(k)}^\bullet$ s.t. $(\eta_p \Omega_{\tilde{X}/W(k)}^\bullet)^j = \left\{ w \in p^j \Omega^j : d(w) \in p^{j+1} \Omega^{j+1} \right\}$

Suppose we also have a lift $F_{W(k)}$ of the relative Frob $F_{X/k} : X \rightarrow X^{(p)}$.

map $F_{W(k)}^* \Omega_{\tilde{X}^{(p)}/W(k)}^\bullet \rightarrow F_{W(k)*} \Omega_{\tilde{X}/W(k)}^\bullet$
 $\uparrow$
 twist by σ
 on $\text{Spa } W(k)$

Lemma $F_{W(k)}^*$ induces a quasi-isom. onto $(\eta_p F_{W(k)*} \Omega_{\tilde{X}/W(k)}^\bullet)^\bullet$

idea. it maps to η_p b/c $d(f^p + pg) = p(f^{p-1}df + dg)$

quasi-isom. b/c mod p , realizes the Cartier isomorphism.

More generally, A gauge is a function $\varepsilon : \mathbb{Z} \rightarrow \mathbb{N}$ s.t. $\varepsilon(i)-1 \leq \varepsilon(i+1) \leq \varepsilon(i)$, $\forall i$

Given a cplx $K^\bullet$ of $W(k)$ -modules, def'n

$$K_\varepsilon^i := p^{\varepsilon(i)} K^i \cap d^{-1} \left(p^{\varepsilon(i+1)} K^{i+1} \right)$$

(can be defined for any function ε)

General form of lemma $\exists$ comm. diagram

ε gauge

$$\begin{array}{ccc}
 [\Omega_{\tilde{X}/W}^\bullet]_\varepsilon & \longrightarrow & \Omega_{\tilde{X}^\sigma/W}^\bullet \\
 \downarrow F^* & & \downarrow F^* \\
 F_* [\Omega_{\tilde{X}/W}^\bullet]_\eta & \xrightarrow{\quad} & F_* \Omega_{\tilde{X}^\sigma/W}^\bullet
 \end{array}$$

where $\eta(i) = \varepsilon(i) + i$

left vertical is quasi-isomorphism.

Applying this to crystalline cohomology

Let $u: (\tilde{X}/W)_{\text{cris}} \rightarrow \tilde{X}_{\text{zar}}$

$J_{X/W}$ PD ideal

$J^{[i]}$ divided power filtration

For any $\varepsilon: \mathbb{Z} \rightarrow \mathbb{N}$, define $J^{[\varepsilon]} = \sum p^{\varepsilon(i)} J^{[i]} \subset J_{X/W}$

$X \hookrightarrow Y$ closed embedding

$\hat{D}$ p-adic completion of PD envelope of X in Y .

$\rightsquigarrow \Omega_{\hat{D}/W}^\bullet$ Hodge filtration $F^\bullet$ $\rightsquigarrow F^{[\varepsilon]} \Omega_{\hat{D}/W}^\bullet$

Th. $\exists$ nat'l q isom $Ru_* J^{[\varepsilon]} \xrightarrow{\sim} F^{[\varepsilon]} \Omega_{\hat{D}/W}^\bullet$

How does this match w/ relative Frobenius?

Th 1 (BO 8.20) If ε tame gauge and $\eta(i) = \max(0, \varepsilon(i) + i)$

Then $\exists$ diag.

$$\begin{array}{ccc}
 Ru_* J^{[E]} & \longrightarrow & Ru_* \mathcal{O}_{\tilde{X}/W} \\
 \downarrow \leftarrow \text{q.isom.} & & \downarrow F^* \\
 F_* \underbrace{L\eta Ru_* \mathcal{O}_{\tilde{X}/W}}_{\text{decalage operator}} & \longrightarrow & F_{\tilde{X}*} Ru_* \mathcal{O}_{\tilde{X}/W} \\
 & & \text{associated w } \eta
 \end{array}$$

How is this used to relate Frobenius & Hodge?

Th (Bo 8.26) X/k smooth proper.

$$\pi_X: H_{\text{cris}}^n(X/W) \xrightarrow{\text{red mod } p} H_{\text{dR}}^n(X/k)$$

$$\Phi: H_{\text{cris}}^n(X^{(p)}/W) \xrightarrow[\text{Frob.}]{\text{rel.}} H_{\text{cris}}^n(X/W)$$

$$\text{Let } M^k = \Phi^{-1}(p^k H_{\text{cris}}^n(X/W)) \subset H_{\text{cris}}^n(X^{(p)}/W)$$

$$\text{Then } \pi_{X^{(p)}} \text{ maps } M^k \text{ onto } F_{\text{Hodge}}^k H_{\text{dR}}^n(X^{(p)}/k)$$

Similar thing for conjugate filtration.

provided that: ① $H_{\text{cris}}^n(X/W)$ torsion free,

② Hodge ss degenerates at E_1 .

Main input is 8.20 applied to several specific tame gauges.

Newton-above-Hodge. weaker than above, but uses the same proof ingredients.